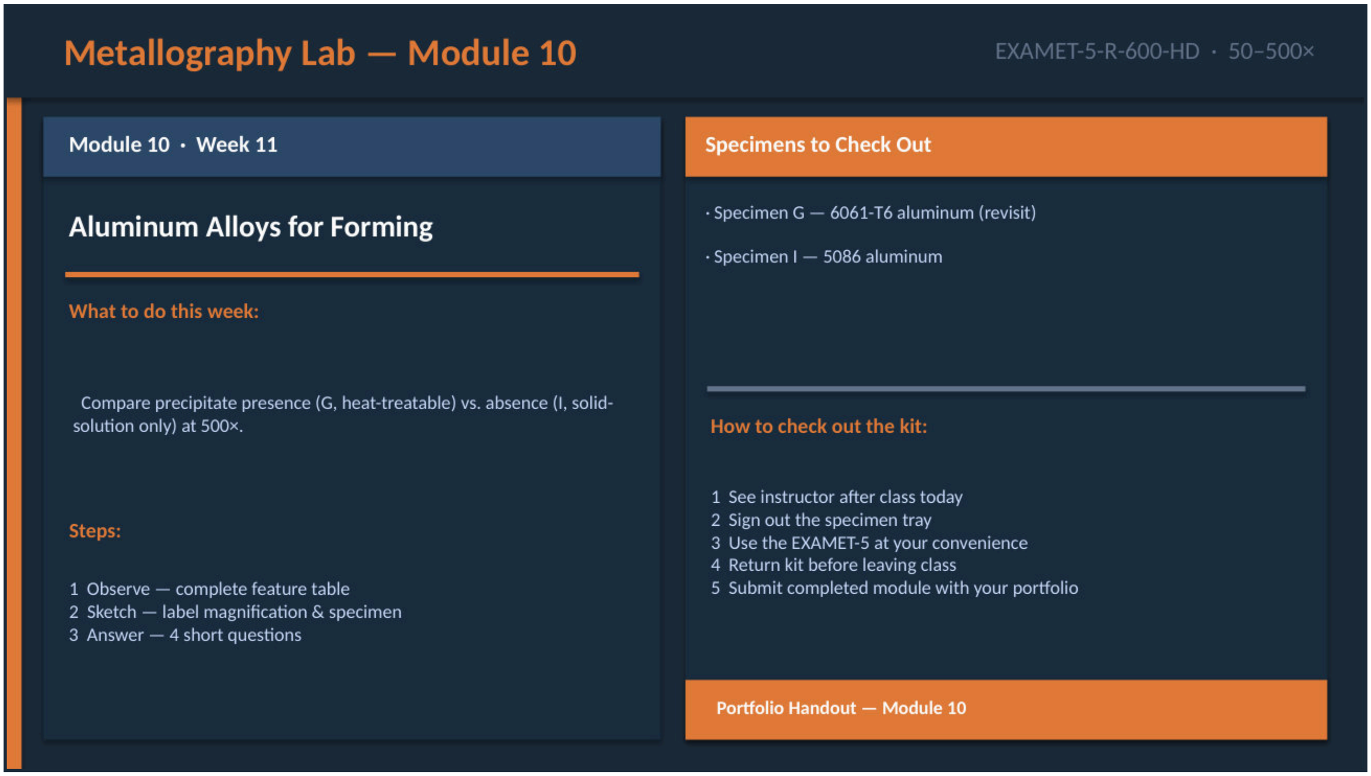
MECH 191 — Week 11
Metal Forming Processes

Name: ____________________ Date: ____________________



SLIDE 1

NOTES

Class Agenda

01 **Hot & Cold Working + Work Hardening**
Recrystallization temperature as the boundary between hot and cold working; how dislocations accumulate, strengthen, and are removed by annealing

02 **Rolling, Forging, Extrusion & Drawing**
The four major forming processes — what each produces, where each is used, and what makes forgings the highest-integrity formed part

03 **Grain Flow & Forming Defects**
Why grain flow gives forgings superior fatigue performance, and the characteristic defects of each forming process and how to detect them

SLIDE 2 *Class Agenda*

NOTES

Metallography Lab — Module 10

EXAMET-5-R-600-HD · 50–500×

Module 10 · Week 11

Aluminum Alloys for Forming

What to do this week:

Compare precipitate presence (H, heat-treatable) vs. absence (I, solid-solution only) at 500×.

Steps:

1 Observe — complete feature table
2 Sketch — label magnification & specimen
3 Answer — 4 short questions

Specimens to Check Out

· Specimen H — 6063 aluminum

· Specimen I — 5052 aluminum

How to check out the kit:

1 See instructor after class today
2 Sign out the specimen tray
3 Use the EXAMET-5 at your convenience
4 Return kit before leaving class
5 Submit completed module with your portfolio

Portfolio Handout — Module 10

SLIDE 3 *Metallography Lab — Module 10 EXAMET-5-R-600-HD · 50–500×*

NOTES

MECH 191 — Course & Unit Roadmap

Unit 1 Materials Fundamentals	Unit 2 Steel & Alloys	Unit 3 Testing & Quality	Unit 4 Manufacturing	Unit 5 Joining & Failure
Wk 1 Aug 24 Atomic Structure & Bonding	**Wk 4** Sep 14 Carbon & Alloy Steels	**Wk 7** Oct 5 Mechanical Testing	**Wk 10** Oct 26 Casting	**Wk 14** Nov 23 Welding Processes
Wk 2 Aug 31 Mechanical Properties	**Wk 5** Sep 21 Stainless & Cast Irons	**Wk 8** Oct 12 NDT Methods	**Wk 11 ← here** Nov 2 Forming	**Wk 15** Nov 30 Weld Metallurgy & Failure
	Wk 6 Sep 28 Nonferrous & Phase Diagrams	**Wk 9** Oct 19 Quality Standards	**Wk 12** Nov 9 Machining	**Wk 16** Dec 7 Corrosion & Wrap-up
			Wk 13 Nov 16 Powder Metallurgy	

Week 17 · FINAL EXAM

SLIDE 4 *MECH 191 — Course & Unit Roadmap*

NOTES

Week 11 — At a Glance

Topics

- Hot working vs. cold working — recrystallization temperature as the boundary; why hot working prevents work hardening and cold working enables it
- Work hardening — dislocation accumulation, anisotropy, strain aging, and the three stages of annealing: recovery, recrystallization, grain growth
- Rolling — flat rolling, shape rolling, hot vs. cold rolling, and characteristic rolling defects
- Forging — open die vs. closed die, flash, grain flow, and why forgings outperform castings and machined bar in fatigue and impact
- Extrusion & drawing — direct vs. indirect extrusion, wire and tube drawing, deep drawing, and the characteristic defects of each
- Forming defects — laps, seams, bursts, pipe defects, delamination, chevron cracking, and NDT detection methods for each

Resources

Reference Material

Kalpakjian Ch. 13, 14 & 15
Rolling · Forging · Extrusion & Drawing

Visualizers

Hot vs Cold Working & Work Hardening
Forging — FIA Design & Grain Flow
Extrusion — AEC Design & Animation
Forming Defects & Process Comparison
Forming Limit Diagrams

Also Covers

Tube & Pipe Drawing — TPA Manufacturing

Next: Week 12 — Machining

SLIDE 5

NOTES

Core Questions

Try to answer these before we dive in — we'll revisit them with answers at the end of class.

1. What is the difference between hot working and cold working, and why does it matter?
2. What is work hardening and how does it affect the strength, ductility, and residual stress of a formed part?
3. What are the main forming processes and what does each one produce?
4. What is grain flow and why does it give a forged part superior performance compared to a casting or machined bar?
5. What defects are unique to forming processes and how is each one detected?

SLIDE 6 *Core Questions*

NOTES

Hot vs. Cold Working — Recrystallization Temperature is the Boundary

Hot Working (above Trex)

No work hardening
Continuous recrystallization annihilates dislocations as fast as they form — large reductions in a single pass without cracking

Lower flow stress
Less force required — enables shaping of large cross-sections (structural plate, rail, ship plate)

Recrystallized grain structure
Final grain size controlled by finishing temperature and reduction; finer finishing = finer grains

Scale formation
Oxidation at high temperature produces mill scale — must be removed (pickling) before cold finishing

Inferior tolerance & finish
Thermal expansion variation and scale give rough surface; ±1–3 mm typical for hot rolled plate

Trex
≈ 0.4Tm

Cold Working (below Trex)

Work hardening
Dislocations accumulate — strength and hardness increase, ductility decreases with each pass; eventually must anneal

Better surface finish
No scale, less thermal distortion — achieves Ra 0.8–3.2 µm and ±0.1 mm dimensional tolerances

Residual stress
Compressive stress at surface (beneficial for fatigue resistance); tensile stress in the interior

Annealing required
When ductility is exhausted, intermediate annealing restores formability through recrystallization

Warm working
Between room temp and Trex — partial recovery without full recrystallization; compromise for Al and some steels

SLIDE 7 *Hot vs. Cold Working — Recrystallization Temperature is the*

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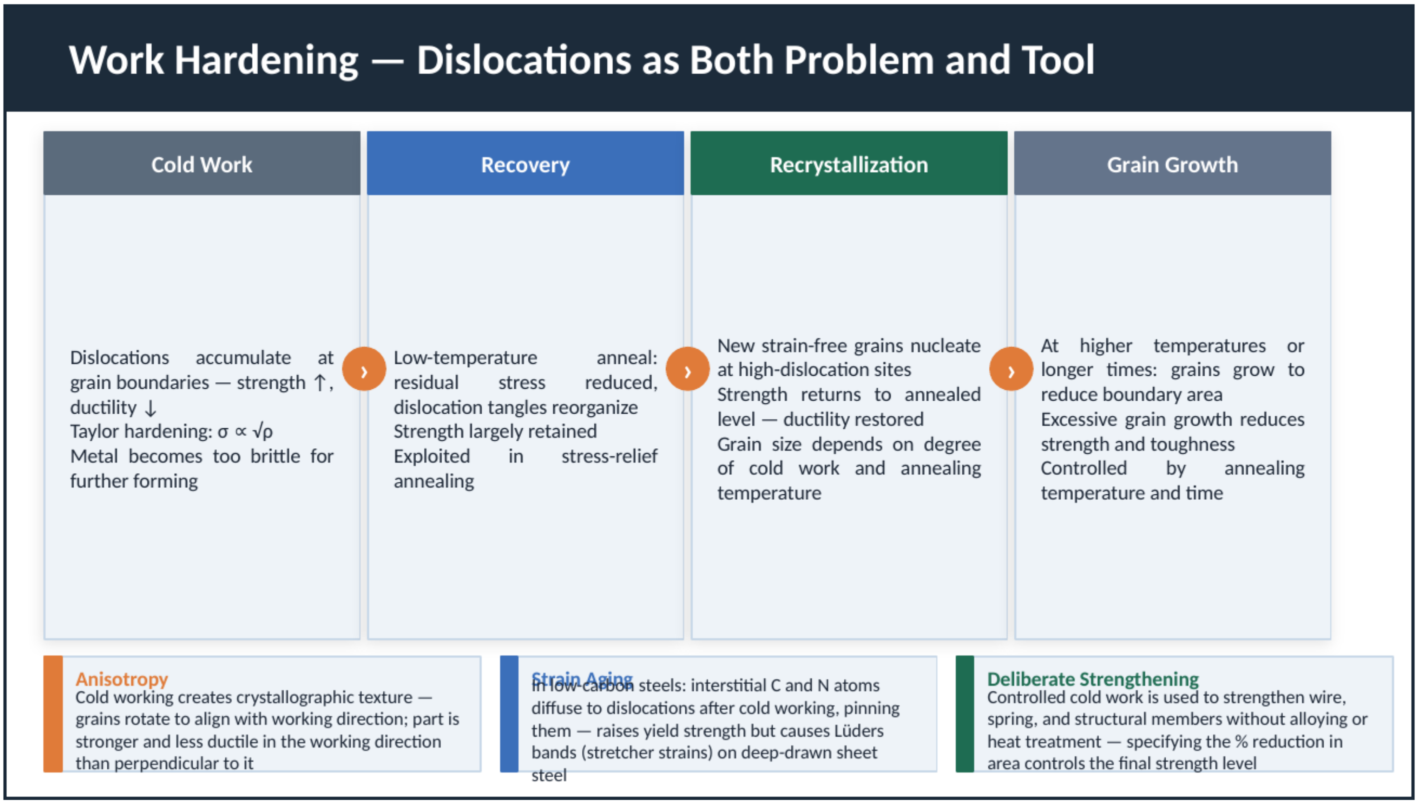

SLIDE 8 *Work Hardening — Dislocations as Both Problem and Tool*

NOTES

Rolling — The Highest-Volume Forming Process

Hot Rolling

- Continuously cast slab reheated above Trex → successive rolling stands reduce thickness progressively
- Finishing temperature controls final grain size — lower finish temp = finer grains = better properties
- Blue-gray mill scale surface; rough tolerance ±1–3 mm; used for structural plate, beam, rail, pipe
- Shape rolling: specially shaped rolls produce I-beams, channels, angles in a single pass sequence

Cold Rolling

- Hot rolled strip pickled (acid cleaned to remove scale) → cold rolled to final gauge
- Bright, smooth surface; tight thickness tolerance ±0.05 mm; strong in rolling direction
- Final batch or continuous anneal restores ductility for deep drawing (automotive panels, cans)
- Ring rolling: thick-walled ring reduced in cross-section and expanded in diameter — seamless bearing rings, flanges with circumferential grain flow

Rolling Defects

Edge Cracking

Edges cool faster in hot rolling → tensile stress at roll exit → edge cracks; insufficient ductility at cooler edges. Detected: VT

Alligatoring

Severe internal cracking during hot rolling of billets with poor ductility — top and bottom surfaces separate like a mouth opening. Detected: VT

Wavy Edges / Center Buckle

Non-uniform thickness across width from roll bending — one edge or the center is thicker, causing flatness defects in sheet. Detected: dimensional

Laps & Seams

Metal folds back on itself at the edges during rolling — linear surface indications parallel to rolling direction; fatigue crack initiation sites. Detected: MT, PT

Pipe (Lamellar Defect)

Ingot shrinkage void drawn out into a central planar defect during primary rolling — dangerous under through-thickness load in welded connections. Detected: UT

SLIDE 9 *Rolling — The Highest-Volume Forming Process*

NOTES

Forging & Grain Flow — Highest-Integrity Forming Process

Open Die Forging

Metal compressed between flat or simple shaped dies — flows freely laterally. Used for very large components (shafts up to 100 tonnes, discs, rings) and for breaking down ingot structure. Requires skilled operator to control shape progression. Produces a worked, refined microstructure throughout the cross-section.

Closed Die (Impression Die) Forging

Metal compressed in a die cavity machined to the part shape — metal fills the cavity and excess escapes as flash at the parting line. Flash builds back-pressure to completely fill the die before escaping; trimmed after forging. Produces near-net-shape with excellent grain flow and dimensional consistency.

Flash: the thin fin squeezed out at the die parting line — a functional feature, not a defect. Its resistance to flow forces complete die fill.

Grain Flow — Why Forgings Win

What grain flow is
Grains elongate and align in the direction of metal flow during forging — a fiber-like structure analogous to wood grain. Revealed by etching a cross-section with acid.

Why it matters
A well-designed die orients grain flow to follow the part contour and resist primary service loads. Grain flow aligned with load path = maximum fatigue and impact resistance.

vs. machined bar
A part machined from bar stock has grain flow running straight through — the grain cuts across the critical geometry. A forging wraps the grain around the shape.

vs. casting
Castings have no grain flow — random, equiaxed grains with no preferred orientation. Forgings outperform equivalent castings in fatigue life by factors of 2–5×.

Laps: the failure mode
If the die causes metal to fold back on itself, the grain flow creates a plane of weakness — a lap. A folded grain flow is always rejectable in fatigue-loaded components.

SLIDE 10 *Forging & Grain Flow — Highest-Integrity Forming Process*

NOTES

Extrusion & Drawing — Continuous and Pulled Product

Direct (Forward) Extrusion

Ram pushes heated billet through die — extrude exits in direction of ram travel. High container friction; most common type. Extrusion ratio 10:1 to 100:1 for aluminum. Produces rods, tubes, angles, and complex constant-cross-section profiles. Al at 400–500°C; steel at 1100–1250°C with glass or graphite lubricant.

Indirect (Backward) Extrusion

Die moves into the stationary billet — no relative movement between billet and container wall. Lower friction, more uniform deformation. Used for precise profiles and difficult alloys (superalloys, titanium). Cold impact extrusion: aluminum, copper, soft steels at room temperature — produces thin-walled tubes and cups.

Wire, Tube & Deep Drawing

- Wire drawing: rod pulled through progressively smaller dies — 15–25% reduction per die; intermediate annealing for large total reductions
- Tube drawing: mandrel controls internal diameter — tight-tolerance seamless tubing for hydraulic and medical applications
- Deep drawing: sheet blank drawn into cup by punch — automotive panels, beverage cans; requires good ductility and appropriate crystallographic texture
- Drawing is compressive die + tensile pull — distinguished from extrusion (compressive only)

Extrusion & Drawing Defects

Pipe (Extrusion): Oxide from billet surface drawn into extrude center near billet end — every extrusion back-end is cropped. UT detects.

Fir Tree Cracking: Periodic transverse surface cracks at die exit — excessive extrusion speed or temperature. VT detects.

Chevron / Center Burst: Internal cracks along centerline — excessive hydrostatic tension in drawing or extrusion deformation zone. UT, RT detects.

SLIDE 11 *Extrusion & Drawing — Continuous and Pulled Product*

NOTES

Forming Defects — Process, Cause, and Detection

Defect	Process	Cause	Location	Detection
Laps	Rolling, Forging	Metal folds back on itself at surface	Surface, parallel to working direction	**MT, PT**
Seams	Rolling	Surface cracks from ingot closed but not welded	Surface, parallel to rolling direction	**MT, PT**
Bursts	Forging, Extrusion	Tensile stress in deformation zone exceeds ductility	Internal, linear	**UT, RT**
Pipe	Rolling, Extrusion	Ingot shrinkage void drawn to center	Central, linear or planar	**UT**
Delamination	Rolling	Pipe or inclusions separate under through-thickness load	Internal, planar	**UT**
Chevron Cracking	Drawing, Extrusion	Hydrostatic tension at centerline	Centerline, internal	**UT, RT**
Orange Peel	Cold Working	Coarse grain deforms non-uniformly	Surface	**VT**

SLIDE 12 *Forming Defects — Process, Cause, and Detection*

NOTES

Technician's Perspective

Knowing the process tells you where to look and what you're looking for.

Red Flag: Rolled plate accepted for a welded T-joint without UT

Plate passes VT and MT — no rejectable indications. It is welded into a T-joint with through-thickness tensile stress. The plate later delaminates at the weld. MT and PT cannot see lamellar pipe defects lying parallel to the plate surface. All rolled plate used in through-thickness-loaded connections must be UT-tested for lamellar defects before fabrication. This is a code requirement, not optional.

Caution: Linear MT indication on a forged crankshaft — orientation matters

MT reveals a linear indication in the crankpin fillet. Before disposition: determine orientation relative to grain flow. Parallel to grain flow = less critical than transverse — a cross-grain crack cuts across fibers rather than following them. Read the etch pattern to establish grain flow direction, then evaluate. Orientation does not eliminate the need to evaluate against acceptance criteria, but it determines urgency and disposition.

Good Practice: Linear PT indications on drawn hydraulic tube — reject, don't size

PT on drawn hydraulic tubing reveals linear indications parallel to the tube axis. Could be seams (ingot surface defects) or laps (metal folding during drawing) — both are rejectable in pressure-rated tubing. The technician does not size them against acceptance limits. Any linear surface indication parallel to the axis concentrates hoop stress from internal pressure and is a fatigue initiation site. Surface quality is part of the process release criteria — not an afterthought.

SLIDE 13 *Technician's Perspective*

NOTES

Core Question Answers

What is the difference between hot working and cold working, and why does it matter?

The boundary is the recrystallization temperature (~0.4Tm). Hot working: continuous recrystallization prevents work hardening — large reductions without cracking, recrystallized equiaxed grain structure, inferior surface finish. Cold working: dislocations accumulate — strength increases, ductility decreases, better surface finish, annealing required when ductility is exhausted.

What is work hardening and how does it affect a formed part?

Plastic deformation accumulates dislocations; as they tangle and pile up, further movement becomes harder — strength and hardness increase, ductility decreases. The part has anisotropic properties (stronger in working direction), residual compressive surface stress (good for fatigue), and loses its work-hardened strength if heated above the recrystallization temperature.

What are the main forming processes and what does each produce?

Rolling: passes slab between rolls — plate, sheet, strip, structural sections. Forging: compressive force through dies — highest-integrity parts (crankshafts, flanges, aircraft structure). Extrusion: forces through die opening — constant cross-section profiles (Al extrusions). Drawing: tensile pull through die — wire, tube, bar at tight tolerances.

What is grain flow and why does it give a forged part superior performance?

Forging elongates and aligns grains in the direction of metal flow — a fiber-like structure. A well-designed die orients grain flow to follow the part contour and resist service loads. This gives forgings 2–5× better fatigue life than equivalent castings (no grain flow) or machined bar (grain flow cuts across the geometry).

What defects are unique to forming processes and how is each detected?

Laps and seams: surface linear defects parallel to working direction — MT, PT. Bursts: internal cracks from excessive tensile stress in forging or extrusion — UT, RT. Pipe: ingot shrinkage void drawn to center of rolled or extruded product — UT. Delamination: pipe or inclusions separating under through-thickness load in rolled plate — UT. Chevron cracking: centerline cracks in drawing/extrusion — UT, RT.

SLIDE 14 *Core Question Answers*

NOTES

Key Takeaways — Week 11: Metal Forming Processes

1 Hot working and cold working produce fundamentally different microstructures and properties — hot working refines grain structure through recrystallization while cold working strengthens through dislocation accumulation; the choice between them determines final mechanical properties as much as the alloy composition does. The boundary is recrystallization temperature, not whether the metal feels hot.

2 Grain flow in forgings is a deliberate microstructural feature that gives forged components superior fatigue and impact resistance compared to castings or machined bar — grain flow should follow the part contour and resist the primary service loads. A technician who can read a grain flow etch pattern can assess whether a forging die was designed correctly, and can recognize a lap as a plane of weakness rather than just a surface mark.

3 Every forming process produces its own characteristic defects — laps and seams in rolling and forging, pipe defects in rolled plate and extrusion, center bursts in drawing, delamination in welded rolled plate connections — and a technician who understands the process can anticipate where to look, what the defect looks like, and which NDT method will find it.

SLIDE 15 *Key Takeaways — Week 11: Metal Forming Processes*

NOTES

Resources & What's Next

Week 11 Resources

Kalpakjian Ch. 13, 14 & 15
Rolling of Metals · Forging · Extrusion & Drawing — primary reading

Hot vs Cold Working Visualizer
Work hardening curve, recrystallization stages, annealing effects — interactive

Forging Visualizer
FIA design guidelines, grain flow animation, open vs. closed die — interactive

Extrusion Visualizer
AEC design guidelines, direct vs. indirect extrusion animation — interactive

Forming Defects Visualizer
Process comparison and defect recognition for each forming process — interactive

Week 12 — Machining

November 9, 2026 · Unit 4: Manufacturing continues

- What happens at the cutting zone — shear plane, chip formation, heat generation, and surface integrity
- Turning, milling, drilling, and grinding — what each produces and what surface finish each achieves
- Cutting tool materials — HSS, cemented carbide, coatings, ceramics, CBN, and PCD
- Surface finish parameters — Ra, Rz, and why surface roughness affects fatigue, sealing, and wear
- Machining defects — built-up edge marks, chatter, grinding burn, and recast layer in EDM

SLIDE 16 *Resources & What's Next*

NOTES